SECTION A : DATA FUNDAMENTALS

1. WHAT IS A DATABASE?

A database is a collection of data that is stored electronically, organized into rows & columns within tables and can easily be managed.

EXPLAIN WHY DATABASES ARE ESSENTIAL IN MODERN WORLD

DATABASES ARE ESSENTIAL BECAUSE THEY HELP STORE LARGE AMOUNTS OF INFORMATION EFFICIENTLY BY POWERING EVERYDAY APPLICATIONS.

PROVIDE TWO REAL-WORLD EXAMPLES OF WHERE DATABASES ARE USED.

BANKING SYSTEMS : INVESTIGATING ACCOUNT TRANSACTIONS

SCHOOL RECORD : MANAGING & STORING STUDENT DATA

2. WHAT IS A DATA WAREHOUSE?

A data warehouse is the central database that stores large amounts of data from different source for analytical processing.

HOW DOES IT DIFFER FROM A REGULAR DATABASE?

It differs as a regular database is used for daily operations, while a data warehouse is used for reporting and analysis trends.

WHY WOULD A COMPANY USE A DATA WAREHOUSE?

Companies would use a data warehouse to make better business decisions and for strategic planning.

3. WHAT IS A DATA LAKE?

DATA LAKE IS LIKE A DUMPING GROUND, WHERE DATA CAN BE STORED RAW, UNSTRUCTURED, SEMI-STRUCTURED AND STRUCTURED.

How does Data Lake differ from a Data Warehouse?

The difference is a data warehouse stores processed data, while a data lake stores raw data that does not have a format and is not consistent.

When would a business prefer a Data Lake over a Data Warehouse?

Businesses would prefer data lake when a business needs flexibility and wants to analyse large and advanced data.
Businesses wanting to store diverse data types and for cost-effective storage purposes with the big data.

4. What is a Data Mart?

A data mart is a smaller section of a data warehouse that focuses on a specific department within the business, e.g., sales.

How does it relate to a Data Warehouse?

DATA MART IS DESIGNED TO PROVIDE ANALYTICAL DATA TO A SPECIFIC FUNCTIONAL OR DEPARTMENT AREA JUST LIKE DATA WAREHOUSE PROVIDES ANALYTICAL DATA FOR THE WHOLE BUSINESS.

THE DATA USED IS RELIABLE, STRUCTURED AND CONSISTENT AS IT IS OBTAINED DIRECTLY FROM A DATA WAREHOUSE.

Who typically uses Data Marts?

Business Employees: Daily operational reporting and projects
Data Analysts: Demanding aimed datasets for specialized, driven analysis
Business Intelligence Teams: Dashboard & reporting purposes

5. EXPLAIN THE DIFFERENCE BETWEEN STRUCTURED, SEMI-STRUCTURED, AND UNSTRUCTURED DATA. PROVIDE ONE EXAMPLE OF EACH.

Structured data: Data that is consistent, factual and organised in tables e.g. spreadsheets and application forms.

Semi-structured data: Data that is partly organised, has partial structure characterstics but does not form part of the relational database, e.g. XML files.

Unstructured data: Data that does not have a consistent format, not organized and has no specific format e.g videos and social media posts.

6. WHAT IS DATA MODELLING?

Data Modelling is the process of designing the life cycle of data from the point of collection, to the storage phase up until the usage phase within the system.

WHY IS IT IMPORTANT WHEN DESIGNING A SYSTEM THAT WORKS WITH DATA?

It is important because the system needs to ensure data is organised by providing consistency and quality, easy to use by providing a common language between users and developers.

DESCRIBE AT LEAST TWO TYPES OF DATA MODELS.

Entity-Relationship (ER) Model: is a high-level overview of data mapping by outlining data specifics using models focusing on business areas and how they connect.

Logical model: shows relationships between entities and structure in detail.

7. WHAT IS DATA MINING?

Data mining is the process of analysing large datasets to discover certain patterns, particular customer behaviors, analysing specific trends and developing useful information.

HOW IS IT USED IN BUSINESS?

- Businesses use data mining to understand customer behavior and increase profits by making informed decisions.
- Businesses also use data mining to analyse and investigate mischievous or unsafe behaviour e.g. huge withdrawals of money from the same bank card.
- Optimizing business processes by identifying certain bottlenecks and system latency by ~~monitoring~~ analysing service and system usage data

PROVIDE TWO REAL-WORLD EXAMPLES OF DATA MINING IN ACTION

- Retail store predicting what customers will buy or might have interest in.
- Banks noticing suspicious activities from a certain account detecting fraud.

8. WHAT IS A DATABASE MANAGEMENT SYSTEM (DBMS)?

A DBMS is a system / software used manipulate, control and manage databases by allowing users to have access to read, delete, update, insert and query database effectively.

BRIEFLY DESCRIBE FOUR POPULAR DBMS

MySQL: Relational database management system owned by Oracle, identified for it's speed, simplicity and ease of use

SUITABLE USE CASE: MySQL is used for websites and applications.

PostgreSQL : Object-relational database management system (ORDBMS) recognized for its consistency, data reliability and precise SQL compliance.

SUITABLE USE CASE : PostgreSQL is used for advanced data analysis.

MongoDB : NoSQL design ~~structured~~ database that stores unstructured ~~data flexible~~ data

SUITABLE USE CASE : Is used for flexible, unstructured data that may change over time

Oracle Database : Multi-model database system used by enterprises recognized for enhanced performance and continuous securing properties

SUITABLE USE CASE : Used for Enterprise Resource Planning, back-office strategies.

9. WHAT IS ETL (EXTRACT, TRANSFORM, LOAD)?

ETL is a data pipeline process designed to move ~~to process~~ raw data from multiple databases by undergoing multiple stages of collecting data, cleaning and organizing data until ~~and~~ storing data.

EXPLAIN EACH STEP

Extract : During this phase, raw data is collected from different sources to a temporary storage area.

Transform : In the temporary storage area the process of cleaning and organising data take places

Load : The cleaned and organised data is then moved into the central database e.g data warehouse.

WHY IS ETL PROCESS IMPORTANT IN DATA ANALYTICS?

ETL is important in data analytics to ensure that data is reliable, clean and structured to prepare for analysis and decision-making.

10. WHAT ARE COMMON DATA STORAGE FORMATS USED IN ANALYTICS AND BRIEFLY DESCRIBE EACH?

(~~Comma~~ Comma-Separated-Values)

CSV: is the most common and used data formats mainly for tabular data. CSV can be opened by simple text formats and spreadsheet functions like Microsoft Excel.

CSV can be used for: Used for small datasets when storing and moving data between different applications e.g from Microsoft Excel CSV to Databricks.

JSON (JavaScript Objection Notation): Format is famous in web and application progress. It is a flexible format that permits data switch between systems.

JSON can be used for: Used to store and substituting data between a server and web application.

Parquet: is a format that stores data using columns for well organized data and high-performance analysis used in big data processing.

Parquet can be used for: Used for storing and controlling big data of any format by using cloud storage space.

Avro: is a format that stores data using rows and is designed to simplify data switch between different languages and processing systems.

Avro can be used for: is used to compact and support schemas used in data application.

SECTION B: ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

11. WHAT IS ARTIFICIAL INTELLIGENCE (AI)?

AI is the ability of machines (computers, robots etc) to perform tasks that normally require human intelligence.

DIFFERENCE BETWEEN NARROW AI (WEAK AI) AND GENERAL AI (STRONG AI). PROVIDE ONE EXAMPLE OF EACH

Narrow AI (Weak AI): is designed for a specific task, and cannot perform any task beside that specific task e.g voice assistants like Siri.

General AI (Strong AI): is aimed to perform any intellectual task like a human

12. WHAT IS MACHINE LEARNING (ML)?

ML is a type of AI where machines learn from data to make decisions and predictions instead of being programmed.

HOW DOES IT DIFFER FROM TRADITIONAL PROGRAMMING?

Traditional programming follows fixated set of rules instructing the machine on what it needs to produce, while ML learns patterns from data.

DESCRIBE THE 3 MAIN TYPES OF MACHINE LEARNING; SUPERVISED, UNSUPERVISED, AND REINFORCEMENT LEARNING.

Supervised: Instructions are learnt from labelled data that already describes interpretations.

Unsupervised: Instructions are learnt from unlabelled data by figuring out patterns.

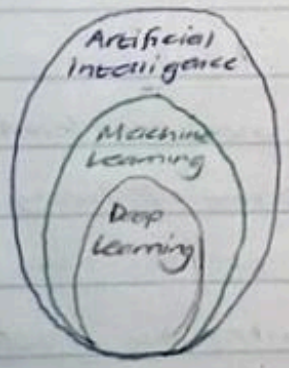
Reinforcement: Depends on trial and error instructions to be able to improve and work on the desired outcome.

10. WHAT IS DEEP LEARNING (DL)?

DL is a type of Machine Learning that changes the way machines accepe, interact and engage by mimicking neutral networks of human brain by discovering certain patterns and making rational decisions from large amounts of data.

HOW DOES IT RELATE TO MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE?

Deep Learning is or forms part of Machine Learning, which Machine Learning also forms part of Artificial Intelligence.



WHAT ARE NEUTRAL NETWORKS AND HOW DO THEY WORK?

Neutral Networks are system structure that mimics how the human brain processes information using layers.

14. WHAT IS NATURAL LANGUAGE PROCESSING (NLP)?

NLP is a field of AI that allows computers to understand and process human language, both ~~writing~~ written and verbally.

PROVIDE THREE REAL-WORLD APPLICATIONS OF NLP THAT YOU USE IN EVERYDAY LIFE.

- Voice assistants like Siri and Alexa
- Chatbots
- Translation apps like Google Translate

15. WHAT IS A LARGE LANGUAGE MODEL (LLM)?

LLM is an advanced AI model trained on large amounts of data by making it possible to understand and generate human language by interpreting texts and summarizing of articles.

HOW IS IT DIFFERENT FROM A STANDARD MACHINE LEARNING MODEL?

LLMs are specialized for language tasks that require a large, unlabeled and unstructured text datasets e.g books, websites and texts, while standard ML models are designed for a specific tasks that require organized, controlled and structured data e.g spreadsheets, SQL and databases.

NAME TWO WELL-KNOWN LLMs AND EXPLAIN WHAT THEY ARE USED FOR

- GPT (Generative Pre-trained Transformer): is mostly used for interactive tasks by following instructions and engaging in conversations like online assistants, customer service and content creation.

- BERT (Bidirectional Encoder Representations from Transformers): is used mainly for search tasks by analysing context within a text like interpreting search query completion and improving search engines.

16. WHAT IS GENERATIVE AI ?

Generative AI creates content such as text, images or music from historic data by analysing patterns.

HOW DOES IT WORK AT A HIGH LEVEL ?

At high level it engages with deep learning model by mimic neutral networks to discover particular patterns by predicting what to follow.

NAME 3 TOOLS THAT USE GENERATIVE AI AND DESCRIBE WHAT EACH ONE DOES

- ChatGPT generates text
- DALLE create images
- Github Copilot generates codes

17. WHAT IS A TRAINING DATASET ?

A training dataset is a dataset used to teach computer models on how to recognize patterns, make predictions using BERT by utilizing the three main types of machine learning.

WHY IS THE QUALITY OF TRAINING DATA SO IMPORTANT WHEN BUILDING AN AI MODEL ?

It is important because high-quality data ensures accurate results by increasing the models ability to master complicated patterns, provide diverse data and ensuring data integrity.

WHAT COULD GO WRONG IF THE TRAINING DATA IS BIASED OR INCORRECT ?

Poor data leads to inaccurate results, project failure, bias, incorrect outcomes and rotational (repetitive) patterns.

18. WHAT IS COMPUTER VISION?

Computer Vision is a field of artificial intelligence that enables machines to interpret and understand visual data like images and videos by converting them into numerical data and using algorithms to detect patterns and objects.

HOW DO MACHINES 'SEE' AND INTERPRET IMAGES?

Machines need to see and interpret images by converting them into numerical data

PROVIDE TWO REAL-WORLD USE CASES OF COMPUTER VISION

- Autonomous vehicles: used to detect road signs, and road users
- Medical Imaging: MRI scans used to detect tumors

SECTION C: APIs, TOOLS AND DEVELOPER CONCEPTS

19. WHAT IS AN API (APPLICATION PROGRAMMING INTERFACE)?

An API collection of guidelines that permits diverse software systems to engage and exchange data efficiently together.

EXPLAIN WHAT IS IT USING A SIMPLE REAL-WORLD ANALOGY.

EXAMPLE IS THE WAITER ANALOGY

- How a waiter transfers requests between a customer and the kitchen, and the response, returns with what ~~that~~ customer has ordered.

WHY ARE APIs IMPORTANT IN MODERN SOFTWARE AND DATA SYSTEMS?

APIs are important because they enable intergration of systems and reuse of services across applications.

20. WHAT IS AN API KEY?

An API key is an identifer used to authenticate and access an API, ensuring that only authorized users can make requests.

WHAT IS ITS PURPOSE?

API serves as an intermediate between different applications to engage and exchange data with one another.

HOW SHOULD IT BE STORED AND PROTECTED, AND WHAT COULD HAPPEN IF IT IS EXPOSED PUBLICLY?

It should be stored securely (e.g., in environment variables) and never exposed publicly, as this ~~as~~ could ~~event~~ lead to unauthorized access and data breaches.

21. WHAT IS A REST API?

A REST API is an ~~architectural~~ architectural style for web service that uses HTTP methods.

WHAT DOES REST STAND FOR?

REST stands for Representational State Transfer

WHAT ARE THE MAIN HTTP METHODS (GET, POST, PUT, DELETE) USED IN REST APIs? BRIEFLY DESCRIBE EACH METHOD.

- GET to retrieve data
- POST to create data
- PUT to update data
- DELETE to remove data

all working together while maintaining stateless communication.

22. WHAT IS JSON (JAVASCRIPT OBJECT NOTATION)?

Is a flexible text-based format mostly used in web and software applications used to store and exchange data switch between systems.

WHY IS IT COMMONLY USED WHEN WORKING WITH APIS AND DATA EXCHANGE?

JSON is a standard practice for API data exchange making it easy for both humans and machines to read and process data.

WRITE A SIMPLE EXAMPLE OF A JSON OBJECT

A JSON object can store user information such as name, age and location in a simple key-value structure.

23. WHAT IS PROMPT ENGINEERING? AN IMPORTANT SKILL WHEN WORKIN WITH AI TOOLS LIKE CHATGPT OR CLAUDE?

Prompt engineering is the practice of writing clear and specific inputs to improve the accuracy and relevance of AI outputs.

WHAT IS PROMPT ENGINEERING?

Prompt engineering is the practice of where planning, evaluating and clarifying of prompts takes place to help generative AI models in achieving accurate, clear and creative responses.

PROVIDE TWO EXAMPLES OF A POORLY WRITTEN PROMPT AND AN IMPROVED VERSION OF EACH

- A vague prompt like "Explain AI" can be improved to "Explain supervised and unsupervised learning in 100 words."

- The Missing Output Format prompt like "Create a summary of this blog" can be improved to "Summarize the following blog into 3 bullet points under the headings: Problem, Solution, and Outcome."

24. WHAT IS A MAP FUNCTION?

MAP function is fundamentals used in programming and mathematics to change data by implementing definite rules or logic to each item on the list.

EXPLAIN WHAT IT DOES IN PROGRAMMING AND DATA PROCESSING

- Programming: It allows codes to be cleaner by frequently replacing for loops.

- Data processing: Mapping matches source data fields to destination fields by ensuring the correct format on the data during the ETL process.

PROVIDE A SIMPLE EXAMPLE IN EITHER PYTHON OR JAVASCRIPT

- Python MAP() with lambda: mapping function that doubles numbers over a list will convert [1, 2, 3] into [2, 4, 6].

25. WHAT IS CLOUD COMPUTING?

Cloud computing is a whenever required availability of computing resources provided as services over the internet by paying for what you will use.

NAME THE 3 MAIN CLOUD SERVICE MODELS (IaaS, PaaS, SaaS) AND BRIEFLY EXPLAIN EACH. GIVE ONE EXAMPLE OF A PRODUCT FOR EACH MODEL

- IaaS: Infrastructure resource used as a server, storage and networking over the internet by paying for what you will use e.g. Amazon Web Services (AWS).
- PaaS: Provides development platforms with full support for building lifecycle and delivering web-based applications e.g. Microsoft Azure App Service.
- SaaS: Provides ready-to-use applications controlled by service providers over the internet. e.g Shopify (e-commerce) and Dropbox (cloud storage).

26. WHAT IS A VERSION CONTROL SYSTEM?

Version Control System is a tool used in software development that tracks controls and documents changes to code over time.

WHAT IS GIT?

Is the version control system that assist you trace and control code change, also allows people to work together on the same project.

WHY IS GIT WIDELY USED BY DATA ANALYSTS AND DEVELOPERS

It allows them to trace changes, work together on certain projects at the same time and providing the ability of safe keeping historic codes securely.

BRIEFLY EXPLAIN WHAT A 'COMMIT' AND 'BRANCH' ARE

- Commit: represents that changes have been saved and processed.

- Branch: allows to isolate independent development path from the main one without affecting the repository.

27. WHAT IS JUPTER NOTEBOOK?

Its a web application designed by allowing users to develop and to form part of records that include live code, equation and visualization.

WHAT PROGRAMMING LANGUAGE IS IT MOSTLY USED WITH?

Primarily used with Python

WHY IS IT POPULAR AMONG DATA ANALYSTS AND DATA SCIENTISTS?

Allows users to combine codes, outputs and explanations in a single document making it ideal for data analysis and reproducible research.

28. WHAT IS PYTHON?

Python is a leading interpreted programming language known for its basic and understandable structure that is usually used by data scientists and AI.

WHY IS IT CONSIDERED ONE OF THE MOST POPULAR PROGRAMMING LANGUAGE FOR DATA ANALYTICS AND AI?.

Python is a leading language in data analytics and programming because of its distinctive fusion of availability, sturdy community support, for its ability to intergrate, rapid running of codes immediately and vast library ecosystem.

NAME 4 PYTHON LIBRARIES COMMONLY USED IN DATA WORK AND BRIEFLY DESCRIBE EACH

- NumPy for numerical computing
- Pandas for data manipulation
- Matplotlib for visualization
- Scikit-learn for machine learning

SECTION D: DATA ETHICS, GOVERNANCE AND CAREERS

29. WHAT IS DATA PRIVACY?

Data privacy is the ~~ug~~ right or protection of individuals to control how their data is collected, managed and accessed by organizations.

WHY IS IT IMPORTANT?

its important as who has the authorization to access the data and how its managed by preventing fraud, identity theft and from attacks. It enables individuals to be aware who is managing and handling their personal information.

RESEARCH THE GDPR (GENERAL DATA PROTECTION REGULATION) AND EXPLAIN WHAT IT IS

GDPR is a large scope of EU privacy law that governs strict regulations and guidelines on managing, processing and recording of personal information of individuals. It provides individuals the ability to ~~have~~ be made aware of who has access to their information, allowing individuals to manage, update, delete and control their information.

WHICH COUNTRIES DOES GDPR APPLIES TO?

United Kingdom and European Union member states, Norway

30. WHAT IS DATA ETHICS?

Data ethics refers to principles directing the responsibility of gathering, security, analysis and utilization of information by ensuring fairness and to avoid harm.

PROVIDE 3 EXAMPLES OF UNETHICAL USE OF DATA OR AI

- collecting personal data of an individual through social media platform and website without their consent.
- The misuse of generative AI to create deepfake to impersonate someone else.
- The use of implementing bias algorithms when hiring based on gender and race.

EXPLAIN WHAT MAKES EACH EXAMPLE PROBLEMATIC

- All the 3 examples mentioned above can lead to discrimination and loss of trust.
- This illustrates that when AI is utilized without the appropriate governance, it can can societal harm.
- Their work against the standard principles of fairness.

31. WHAT IS DATA GOVERNANCE?

Data Governance is a framework of compliance guideline of procedures, roles and technology used to control an organizations data by ensuring consistency, safety and accessibility throughout its lifecycle.

EXPLAIN THE KEY COMPONENTS OF A GOOD DATA GOVERNANCE

- Policies, processes and standards: Data governance policies ensures that the processes are outlined thoroughly according to the policies standards and the workflow of data is followed.

- Access control and role-based permissions: Data governance guide outlines the limitation of access control to critical data. Individuals in an organization should only have access to data they dealing with at a certain point to protect confidential information, limit exposure and secure data use.

- Metadata and lineage management: Assists in enhancing exposure of data across data lifecycle, making auditing or debugging capable.

32. WHAT IS BIAS IN AI?

Bias in AI occurs when models produce unfair or skewed outcomes reflecting discrimination, gender inequality due to biased training data or design.

EXPLAIN WHAT IT MEANS FOR AN AI MODEL TO BE BIASED.

- Data Bias: Facial recognition systems performing poorly on certain demographic groups, which can result in discrimination, unfairness and unethical concerns.

- Algorithmic Bias: Using algorithms in hiring tools to penalize CVs that include specific gender or certain words.

- Human / Cognitive Bias: The labeling of data and prioritization of features deemed to be important during early stages of development.

PROVIDE A REAL-WORLD EXAMPLE OF AI BIAS

- Hiring: Job application devices penalizing CVs not containing certain words on the job description.

DESCRIBE THE POTENTIAL CONSEQUENCES

- Erosion of Trust: Creates a mistrust among excluded groups including people of colour and LGBTQ community.

- Harm to individuals: High interest rates on certain ethnic groups, denied loans and unfair arrests.

33. WHAT ARE THE KEY DIFFERENCES BETWEEN A DATA ANALYST, A DATA SCIENCE, A DATA ENGINEER, AND AN AI ENGINEER? CREATE A SIMPLE TABLE COMPARING THEIR ROLES, SKILLS AND TOOLS

ASPECTS	DATA ANALYST	DATA SCIENCE	DATA ENGINEER	AI ENGINEER
ROLES	Interpreting data to answer questions business questions to retrieve, clean aggregate and process data.	Discover hidden insights and building predictive models	Design, develop and maintain data pipelines and infrastructure	Designs and deploys AI systems using AI framework
SKILLS	SQL, Microsoft Excel, Data Visualization,	Advance statistics, ML Algorithms, Python/R	ETL, Data Modeling, Big data technologies	Deep learning, Neutral Networks, API Development
TOOLS	Tableau, PowerBi, Microsoft Excel, SQL, Databricks Python, Snowflake	Python/R, Scikit-learn, Jupter, Spark	Spark, Airflow, Snowflake, kafka	TensorFlow, PyTorch, Kubernetes, Docker

34. WHAT IS BUSINESS INTELLIGENCE (BI)?

An innovative process used to analyse data, presenting data through visualize like graphs and creating dashboards, providing reports and allowing other individuals to access data.

NAME TWO POPULAR BI TOOLS AND EXPLAIN HOW THEY HELP BUSINESSES MAKE DATA DRIVEN DECISIONS

- Power BI: Platform that enables users to have interactive visualization of data and sharing of data insights across organization.

- Tableau: Software that allows organizations to create dashboards from processed data and data insights for data-driven strategies.

SECTION E: PERSONAL REFLECTION

35. WHICH CONCEPTS FROM THIS ASSIGNMENT DID YOU FIND MOST INTERESTING OR SURPRISING? EXPLAIN WHY.

- One of the concepts I found to be interesting was data mart. A user is able to produce and provide analytical data and insights for a specific unit through data mart.
- Second concept was data mining. The ability of using large dataset, new and old to analyse data and discover certain traits, patterns, and certain behavior to make predictions.

36. WHICH CONCEPT DID YOU FIND MOST DIFFICULT TO UNDERSTAND? WHAT RESOURCES DID YOU USE TO HELP YOU UNDERSTAND BETTER?

- Deep Learning Concept was one of the difficult concepts to understand. Honestly not even sure I am able to explain nor understand deep learning. Only thing I took from the concept was that it mimics the human brain with the neutral networks. The diagram made me have some little clue of the concept which I found on the geeksforgeeks website. geeksforgeeks.org/deep-learning/introduction-deep-learning/

37. HOW DO YOU SEE DATA AND AI AFFECTING YOUR FUTURE CAREER OR EVERYDAY LIFE? BE SPECIFIC WITH EXAMPLES.

DATA AND AI AFFECTING FUTURE CAREER

A software or system may be developed using General AI (Strong AI) where data analysts insert information or write a sentence on a computer requesting an insight it wants. The system will perform data cleaning, data processing, and data aggregation to produce the visuals and dashboards requested.

DATA AND AI AFFECTING EVERYDAY LIFE

As an individual who does not like going to malls the ability of being able to order food, buy clothes and groceries online has made life much easy. Being able to do thing at the comfort of my home has been amazing.

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